

of the queue  $L_q^{l,c1}$ . The total delay incurred to the passenger vehicles is equal to the area under the queue profile. Hence, the total general traffic delay of the lane group  $l$  for cycle  $c_1$  is:

$$d_g^{l,c1} = \frac{L_q^{l,c1} \times r_l^{c1}}{2} \quad (7)$$

### 3.3.2 | Case B: Oversaturated condition

The oversaturated case occurs when the available green time is less than the minimum time required for serving the full length of the queue:

$$g_{qd}^{l,c0} \geq g_l^{c0} \quad (8)$$

The passenger vehicles that have not been cleared in the previous cycle form an initial residual queue that reshapes the queue profile. The queue formation speed of the residual queue is equal to  $v_{qd}$ . Using trigonometry, the time interval between the start of red time and the end of residual queue formation  $t_{rq}^{l,c1}$  can be calculated as:

$$t_{rq}^{l,c1} = \frac{(r_l^{c0} + g_l^{c0}) \times v_{qd} + \left(r_l^{c0} + \frac{L_q^{l,c0}}{v_{qd}}\right) \times (v_{qf}^{l,c1} + v_f)}{(v_f + v_{qd}) - (r_l^{c0} + g_l^{c0})} \quad (9)$$

So the length of the residual queue  $L_{rq}^{l,c1}$  can be expressed as:

$$L_{rq}^{l,c1} = t_{rq}^{l,c1} \times v_{qd} \quad (10)$$

According to the defined parameters, the total queue length for the cycle  $c_1$  consisting of both the residual queue and the normal queue formed after  $t_{rq}^{l,c1}$  is:

$$L_q^{l,c1} = \frac{v_{qd} \times \left(t_{rq}^{l,c1} \times (v_{qd} - v_{qf}^{l,c1}) + r_l^{c1} \times v_{qf}^{l,c1}\right)}{v_{qd} - v_{qf}^{l,c1}} \quad (11)$$

Therefore, the total general traffic delay of the lane group  $l$  for the oversaturated condition can be expressed as:

$$d_g^{l,c1} = L_{rq}^{l,c1} \times r_l^{c1} + \frac{(L_q^{l,c1} - L_{rq}^{l,c1}) \times r_l^{c1}}{2} \quad (12)$$

Since the results of the queue analyses for each cycle is saved to the KATC database, the queuing process of the second cycle  $c_2$  and the consecutive cycles can be similarly analysed based on the signal timing and queue status of the cycle  $c_1$ , obtained from the database.

### 3.4 | Public transport vehicle delay

This section describes the next part of kinematic wave theory analysis which is developed to estimate the delays experienced by PT vehicles in the influence area of intersection. Signal timing, the existence of near-sided PT stops and road-space priority measures (RSP) are among the most influential factors in PT vehicle delay [41]. Moreover, Wu et al. indicated that PT vehicle headway, dwell time at the stop, and the use of RSP measures significantly affect the performance of TSP strategies [42]. Figure 4 illustrates the four cases that the PT vehicle would encounter at the intersection according to the initial position at the start of the first cycle  $x_{0b}^{l,c1}$ . For simplicity, the PT vehicle is assumed to travel with the free flow speed of general traffic  $v_f$ . Moreover, regarding the typical free flow speed of urban networks and the values of cycle lengths, it is assumed that the PT vehicle detected in the intersection influence area would ultimately leave the intersection in two cycles' time. The possible cases are investigated as follows:

#### 3.4.1 | Case A: Undersaturated first and second cycles

- Case A1

In this case, the PT vehicle initial position lies in a range that causes the PT vehicle to join the queue of  $c_1$  :

$$x_{0b}^{l,c1} < v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \quad (13)$$

The time interval between the start of red time and when the PT vehicle joins the queue  $t_b^{l,c1}$  is:

$$t_b^{l,c1} = \frac{x_{0b}^{l,c1}}{(v_{qf}^{l,c1} + v_f)} \quad (14)$$

So the PT vehicle position at the time of joining the queue  $x_b^{l,c1}$  can be expressed as:

$$x_b^{l,c1} = v_{qf}^{l,c1} \times t_b^{l,c1} \quad (15)$$

Hence, from trigonometry, the PT vehicle delay can be calculated as:

$$d_b^{l,c1} = \frac{r_l^{c1} \times (L_q^{l,c1} - x_b^{l,c1})}{L_q^{l,c1}} \quad (16)$$

- Case A2

If the PT vehicle is distant enough from the intersection at the start of red time that it could pass through the intersection without being impeded by the queue:

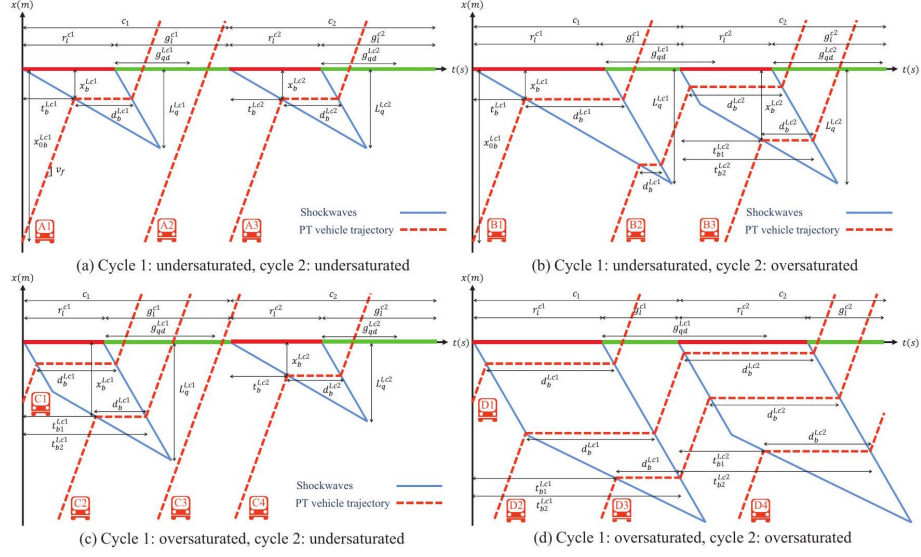


FIGURE 4 Trajectory analysis of PT vehicle based on kinematic wave theory

$$v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \leq x_{ob}^{l,c1} < v_f \times (r_l^{c1} + g_l^{c1}) \quad (17)$$

then the amount of the PT vehicle delay  $d_b^{l,c1}$  would be equal to zero.

- Case A3

In this case, the PT vehicle joins the queue formed in the cycle  $c_2$  in the undersaturated condition similar to the first cycle. The initial position of the PT vehicle can be expressed as:

$$v_f \times (r_l^{c1} + g_l^{c1}) \leq x_{ob}^{l,c1} < v_f \times (r_l^{c1} + g_l^{c1} + r_l^{c2} + g_{qd}^{l,c2}) \quad (18)$$

where  $g_{qd}^{l,c2}$  is the required green time in the cycle  $c_2$  for queue dissipation. Thus, the time elapsed from the start of red time in the second cycle to the point when the PT vehicle joins the

$$t_b^{l,c2} = \frac{x_{ob}^{l,c1} + v_{qf}^{l,c2} \times (r_l^{c1} + g_l^{c1})}{(v_{qf}^{l,c2} + v_f)} - (r_l^{c1} + g_l^{c1}) \quad (19)$$

By obtaining  $t_b^{l,c2}$ , the distance of the PT vehicle from the intersection at the time of joining the queue is:

$$x_b^{l,c2} = v_{qf}^{l,c2} \times t_b^{l,c2} \quad (20)$$

and the PT vehicle delay can be similarly derived from case A1 for the undersaturated queue as:

$$d_b^{l,c2} = \frac{r_l^{c2} \times (L_q^{l,c2} - x_b^{l,c2})}{L_q^{l,c2}} \quad (21)$$

### 3.4.2 | Case B: Undersaturated first cycle and oversaturated second cycle

#### - Case B1

This case describes the situation where the PT vehicle is delayed in the first cycle but the green time is sufficient to pass the PT vehicle through the intersection. The situation can be interpreted as:

$$x_{ob}^{l,c1} < \frac{g_l^{c1} \times v_f \times v_{qd} \times (v_f + v_{qf}^{l,c1})}{v_{qf}^{l,c1} \times (v_{qd} + v_f)} \quad (22)$$

and the PT vehicle delay can be similarly derived from Equation (16).

#### • Case B2

This case is identical to case B1 with a slight difference that the green time of the first cycle  $g_l^{c1}$  is not sufficient to serve the PT vehicle. The position of the PT vehicle can be expressed as:

$$\frac{g_l^{c1} \times v_f \times v_{qd} \times (v_f + v_{qf}^{l,c1})}{v_{qf}^{l,c1} \times (v_{qd} + v_f)} \leq x_{ob}^{l,c1} < v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \quad (23)$$

In this case, the PT vehicle would undergo an additional delay equal to the red time of the second cycle  $r_l^{c2}$ . Thus, the total delay of the PT vehicle constitutes two parts:

$$d_b^{l,c} = d_b^{l,c1} + d_b^{l,c2} = \frac{r_l^{c1} \times (L_q^{l,c1} - x_b^{l,c1})}{L_q^{l,c1}} + r_l^{c2} \quad (24)$$

#### • Case B3

If the initial position of the PT vehicle lies in the following interval:

$$v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \leq x_{ob}^{l,c1} < v_f \times (r_l^{c1} + g_l^{c1} + r_l^{c2} + g_{qd}^{l,c2}) \quad (25)$$

the PT vehicle is solely stopped by the queue of the second cycle. The PT vehicle joins and leaves the queue at time points  $t_{b1}^{l,c2}$  and  $t_{b2}^{l,c2}$ , respectively. The first item can be obtained by intersecting the queue formation shockwave and PT vehicle trajectory as follows:

$$t_{b1}^{l,c2} = \frac{v_{qf}^{l,c2} \times (r_l^{c1} + g_l^{c1} + t_{rq}^{l,c2}) + x_{0b}^{l,c1} - L_{rq}^{l,c2}}{(v_{qf}^{l,c2} + v_f)} - (r_l^{c1} + g_l^{c1}). \quad (26)$$

Hence, the position where the PT vehicle joins the queue  $x_b^{l,c2}$  can be expressed as:

$$x_b^{l,c2} = x_{0b}^{l,c1} - v_f \times (r_l^{c1} + g_l^{c1} + t_{b1}^{l,c2}) \quad (27)$$

and accordingly, the time point for the PT vehicle leaving the queue is equal to:

$$t_{b2}^{l,c2} = r_l^{c2} + \frac{x_b^{l,c2}}{v_{qd}} \quad (28)$$

By obtaining the two mentioned expressions, the PT vehicle delay can be simply calculated as:

$$d_b^{l,c2} = t_{b2}^{l,c2} - t_{b1}^{l,c2} \quad (29)$$

### 3.4.3 | Case C: Oversaturated first cycle and undersaturated second cycle

The first cycle in these cases has a residual queue but the provided green time is capable of serving both the residual and normal queues.

- Case C1

If the PT vehicle joins the residual queue of the first cycle denoted by:

$$x_{0b}^{l,c1} < (v_f + v_{qd}) \times t_{rq}^{l,c1} \quad (30)$$

the PT vehicle delay would be equal to the red time of the first cycle:

$$d_b^{l,c1} = r_l^{c1} \quad (31)$$

- Case C2

This case indicates the situation that the PT vehicle joins the normal queue of the first cycle and is served as the signal indication turns green. The initial position interval can be derived as:

$$(v_f + v_{qd}) \times t_{rq}^{l,c1} \leq x_{0b}^{l,c1} < v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \quad (32)$$

The time points of joining and leaving the queue and the position of the PT vehicle in the queue can be calculated as follows:

$$t_{b1}^{l,c1} = \frac{v_{qf}^{l,c1} \times t_{rq}^{l,c1} + x_{0b}^{l,c1} - L_{rq}^{l,c1}}{(v_{qf}^{l,c1} + v_f)} \quad (33)$$

$$x_b^{l,c1} = x_{0b}^{l,c1} - t_{b1}^{l,c1} \times v_f \quad (34)$$

$$t_{b2}^{l,c1} = r_l^{c1} + \frac{x_b^{l,c1}}{v_{qd}} \quad (35)$$

Consequently, the PT vehicle delay is equal to the time interval between  $t_{b1}^{l,c1}$  and  $t_{b2}^{l,c1}$  :

$$d_b^{l,c1} = t_{b2}^{l,c1} - t_{b1}^{l,c1} \quad (36)$$

- Case C3

The PT vehicle would pass through the intersection without experiencing any delay if:

$$v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \leq x_{0b}^{l,c1} < v_f \times (r_l^{c1} + g_l^{c1}) \quad (37)$$

and therefore, the amount of  $d_b^{l,c1}$  would become equal to zero.

- Case C4

This case is exactly similar to case A3 under which the PT vehicle arrives at the intersection within the second cycle. Therefore, Equations (19)-(21) can be used to estimate the PT vehicle delay.

### 3.4.4 | Case D: Oversaturated first and second cycles

The final case implicitly represents extreme demand levels during peak hours in central urban networks. The PT vehicle trajectories are analysed under the assumption that residual queues are present at the intersection for both the first and the second cycles.

- Case D1

In this case, the PT vehicle reaches the residual queue of the first cycle and is served during the first green time. The related position interval can be expressed as:

$$x_{0b}^{l,c1} < v_f \times g_l^{c1} \quad (38)$$

and the PT vehicle delay is equal to the red time of the first cycle  $r_l^{c1}$ , similar to case C1.

- Case D2

The PT vehicle in the preceding case that could not pass through the intersection during the  $g_l^{c1}$  would have to wait for the next green indication in the second cycle. This situation occurs if:

$$v_f \times g_l^{c1} \leq x_{0b}^{l,c1} < (v_f + v_{qd}) \times t_{rq}^{l,c1} \quad (39)$$

and the resulting PT vehicle delay would constitute both the first and second cycle red times:

$$d_b^{l,c} = d_b^{l,c1} + d_b^{l,c2} = r_l^{c1} + r_l^{c2} \quad (40)$$

- Case D3

If the PT vehicle joins the normal queue of the first cycle which is expressed by:

$$(v_f + v_{qd}) \times t_{rq}^{l,c1} \leq x_{0b}^{l,c1} < v_f \times (r_l^{c1} + g_{qd}^{l,c1}) \quad (41)$$

This case would be almost similar to case C2 with an additional delay item equal to the second cycle red time  $r_l^{c2}$ . Therefore, Equations (33)-(36) are used to determine the PT vehicle delay of the first cycle  $d_b^{l,c1}$  and:

$$d_b^{l,c} = d_b^{l,c1} + d_b^{l,c2} = t_{b2}^{l,c1} - t_{b1}^{l,c1} + r_l^{c2} \quad (42)$$

- Case D4

The final case occurs when the PT vehicle joins the normal queue of the second cycle which is indeed identical to case B3.

### 3.5 | Average passenger delay

The passenger-based nature of KATC requires the total delay experienced by both modes of traffic to be averaged across the entire decision horizon. Hence, the expression for the average passenger delay  $d_a$  within the decision horizon can be presented as follows:

$$\begin{aligned} d_a &= \frac{\sum_{p \in P} d_p}{n_p} \\ &= \frac{\sum_{c \in C} \sum_{l \in L} (o_g \times k_j \times d_g^{l,c} + o_b \times d_b^{l,c})}{\sum_{c \in C} \sum_{l \in L} (o_g \times q_g^{l,c} \times \Gamma + o_b \times n_b^{l,c})} \end{aligned} \quad (43)$$

where  $d_p$  is the delay experienced by each of the passengers,  $\Gamma$  is the cycle length, and  $n_b^{l,c}$  is the total number of PT vehicles detected on the lane group  $l$  in the cycle  $c$ .

### 3.6 | Phase sequence

The mathematical formulation of the phase sequence in the KATC algorithm is developed based on NEMA dual-ring phase structure [43] for a four-legged intersection, indicated in Figure 5. The left-turn movements are assumed to operate in